MaMoRetarget: Morphology-Aware Human-to-Robot Motion Retargeting Across Humanoid Embodiments

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Abstract

Human-to-robot motion retargeting can be viewed as a structured cross-modal motion translation problem, where the semantic intent and temporal dynamics of human motion must be preserved, while respecting embodiment-specific topology, joint constraints, and physical geometry. Existing optimization-based methods can provide geometrically plausible initial mappings, but they are typically per-sequence and non-amortized, and often suffer from self-collisions, unsafe near-contacts, and local temporal artifacts. In this paper, we propose a two-stage framework for motion retargeting across diverse humanoid embodiments. In the first stage, we introduce a minimum-change refinement procedure that repairs collision-prone local kinematic chains through discrete local search. This produces physically improved supervisory trajectories while preserving motion semantics and temporal continuity. We then propose MaMoRetarget, a morphology-aware neural retargeting network that learns an end-to-end feed-forward mapping from structured human motion and robot specifications to executable robot motion. MaMoRetarget integrates a morphology-conditioned human motion encoder, a kinematic tree-based robot specification encoder, and a cross-attention decoder with explicit joint-limit handling, enabling joint modeling of human dynamics and robot-specific embodiment constraints. Experiments on a MoCap-based multi-robot benchmark show that our framework improves physical plausibility and temporal smoothness, supports downstream policy learning, and generalizes well across diverse humanoid embodiments. These results demonstrate that MaMoRetarget offers an efficient and scalable paradigm for high-quality humanoid motion retargeting.

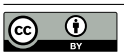
CCS Concepts

• **Computer systems organization** → **Robotic control**; • **Computing methodologies** → *Neural networks*; Motion capture.

Keywords

Human-to-robot Motion Retargeting, Humanoid Robots, Morphology-Aware Learning

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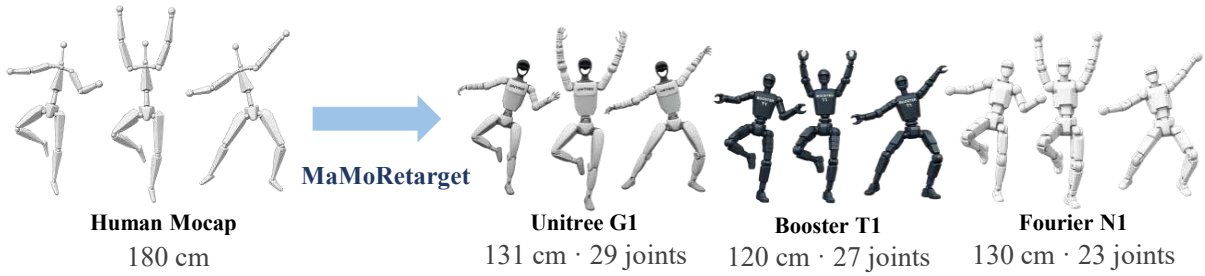


Figure 1: Our method enables the retargeting of human motions to a diverse range of humanoid robot embodiments.

1 Introduction

With the rapid development of Embodied AI, the cross-modal transfer of human motion to robot embodiments is emerging as a critical research problem at the intersection of robotics and multimedia [11, 14, 34]. Human-to-robot motion retargeting is not merely about "pose replication"; rather, it constitutes a highly challenging cross-modal embodied motion translation problem. As illustrated in Fig. 1, a single human motion must be translated into joint trajectories for robots with substantially different scales, DoFs, and kinematic topologies, while preserving its intent, rhythm, and coordinated structure. Due to the structural discrepancies between human and robot embodiments, motions that are natural and feasible for a human often become distorted, physically inconsistent, or even infeasible when directly mapped to robotic systems.

Recent advances in vision-based human motion capture have made it possible to extract motion from monocular videos for downstream robot learning [26, 37]. However, for high-fidelity humanoid retargeting, video-based pipelines introduce additional uncertainty before the retargeting stage even begins. Errors arising from viewpoint changes, occlusions, depth ambiguity, and camera motion can propagate through the pipeline [13]. How to accurately translate clean human motion into executable motion across diverse robot embodiments becomes the core challenge. To tackle this, we focus on a motion-capture (MoCap) setting, where accurate and temporally stable human motion is available, allowing us to study cross-embodiment motion translation in a more controlled and principled manner.

Within this setting, optimization-based retargeting methods [3, 19] serve as a natural starting point for obtaining high-quality retargeting results. They leverage geometric correspondences and kinematic priors to provide robust initial solutions for cross-morphology mapping. However, these methods remain inherently per-sequence and non-amortized. Particularly in a multi-robot setting, our main objective is not merely to "solve for a specific trajectory for a given robot," but rather to learn the shared principles of motion translation across different actions and embodiments. Continuous per-sample optimization is computationally expensive and struggles to meet the real-time processing requirements of large-scale data. More importantly, while the initial mappings generated by these methods may appear geometrically plausible, they are constrained by purely kinematic rules and often yield severe local self-collisions, hazardous near-contacts, and fine-grained temporal artifacts.

To address these limitations, we first propose a minimum-change refinement mechanism. By explicitly repairing collision-related local kinematic chains, this process eliminates local physical defects while preserving the original motion semantics and temporal continuity to the greatest extent possible, thereby constructing high-quality trajectories suitable for use as supervision signals. This process produces physically improved trajectories that are suitable as supervision signals. However, the refinement itself remains sequential and computationally demanding.

Building upon this observation, this paper presents a two-stage motion retargeting framework. The first stage performs offline minimum-change refinement to repair local physical defects and construct high-quality supervisory trajectories. The second stage is MaMoRetarget, a morphology-aware neural retargeting network that distills the geometric and physical priors embedded in the refined trajectories into a fast, unified, and generalizable feed-forward predictor. In this manner, complex kinematic constraints and refinement optimization are utilized offline to construct high-quality supervision, while online inference is performed directly by the reusable MaMoRetarget network

In summary, the main contributions of this paper are as follows:

- We formulate MoCap-based multi-robot retargeting as amortizable cross-embodiment motion translation, establishing a unified learning paradigm for diverse humanoid embodiments.
- We introduce a minimum-change refinement that repairs collision-related kinematic chains, reduces self-collisions and hazardous near-contacts, and retains motion semantics and continuity to produce high-quality supervision.
- We design MaMoRetarget, a morphology-aware network that distills offline geometric and physical priors into a fast, unified, and generalizable feed-forward predictor.
- Extensive experiments demonstrate effectiveness and generalization; we also open-source the code and constructed datasets. Our code has been released at https://github.com/nubot-nudt/MaMoRetarget.

2 Related Work

Motion retargeting originated in computer graphics. Early optimization-based methods used skeletal correspondence, kinematic constraints, and pose consistency [20, 24, 28]; later data-driven approaches learned from paired motions [12, 30], or semantic and structural priors [1, 9, 32]. Recent work covers multi-character interactions [10,

16, 41]and varying body shapes [17, 31, 39]. These methods, however, target animated characters and generally omit the joint constraints, physical feasibility, and contact stability required by robots.

In contrast, motion retargeting for humanoid robots is subject to much stricter structural and physical constraints. The most direct approaches [4, 6]simply map human joint rotations into the target robot joint space, but they ignore the substantial topological and morphological discrepancies between humans and robots. Another line of work [5, 22]formulates whole-body geometric retargeting as an inverse kinematics(IK) problem, matching the positions or orientations of key links in Cartesian space. While effective to some extent, such methods often struggle to fully handle size mismatch and structural inconsistency. Inspired by computer graphics, H2O [8] uses SMPL [18] for scale adaptation before iterative IK; we refer to this family as PHC-style [19]. Such reliance on body models and iterative optimization limits morphology coverage and real-time use. ProtoMotions [29] applies global axis-aligned scaling, whereas GMR [3] further improves retargeting quality through non-uniform local scaling and two-stage IK optimization. Nevertheless, these methods remain fundamentally frame-wise optimization pipelines.

In recent years, data-driven methods have achieved remarkable progress in imitation-learning-based humanoid control [2, 4, 6, 8, 15, 21, 25, 33, 38, 40], yet learning-based full-body retargeting across humanoid embodiments remains limited. Existing methods [27, 35, 36] mainly address simpler arm or upper-body motions and do not fully model cross-robot topology, joint limits, and physical properties. A unified, morphology-aware, physically plausible, and efficient retargeter therefore remains an open problem.

3 Problem Formulation

Human-to-robot motion retargeting is fundamentally modeled as a cross-modal embodied motion translation problem. Given a MoCap sequence, we learn a cross-modal mapping to physically executable motions for diverse robot embodiments while preserving source semantics and temporal characteristics.

3.1 Input Representation

Let $\mathcal{X} = \{\mathbf{X}_{\text{joint}}, \mathbf{X}_{\text{root}}, \mathbf{m}_h\}$ denote an input human motion clip:

- $\mathbf{X}_{\text{joint}} \in \mathbb{R}^{L\times N_h\times D_j}$ contains structured, joint-level kinematic features, where L is the motion clip length. $N_h = 14$ represents the number of human joints involved in the retargeting process. $D_j = 12$, it encompasses local position (3), 6D rotation representation (6), and joint velocity (3).
- $\mathbf{X}_{\text{root}} \in \mathbb{R}^{L\times D_r}$ represents the global root features. A typical $D_r = 12$ configuration includes the root position (3), 6D rotation (6), and root velocity (3).
- $\mathbf{m}_h \in \mathbb{R}^M$ is a human morphological descriptor comprising M elements (e.g., bone lengths).

3.2 Morphology-Aware Robot Embodiment

Each target robot r is defined by a morphological specification

$$\mathcal{S}_r = \{\mathcal{T}_r, \mathbf{a}_r, \mathbf{q}_{\min}, \mathbf{q}_{\max}, \mathbf{l}_r, \mathbf{q}_{\text{rest}}\}, \tag{1}$$

where $\mathcal{T}_r$ denotes the kinematic-topological structure of the robot, including the parent-child relations of the kinematic tree and the discrete joint types; $\mathbf{a}_r$ denotes the local joint axes; $(\mathbf{q}_{\min}, \mathbf{q}_{\max})$ defines the joint limits; $\mathbf{l}_r$ encapsulates link-related geometric properties such as offsets and link geometry descriptors; and $\mathbf{q}_{\text{rest}}$ is the canonical rest pose. Although different robots vary substantially in their degrees of freedom and physical scales, our framework enforces a shared human feature space, enabling a unified multi-robot learning paradigm.

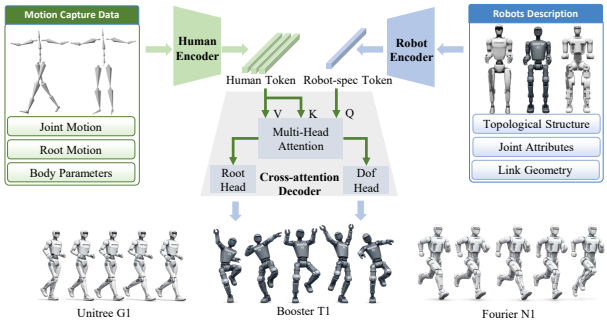


Figure 2: Structure of MaMoRetarget.

3.3 Learning Objective

The objective of this framework is to learn an amortized neural network f_θ that maps the source human motion $\mathcal{X}$ and the target robot specification $\mathcal{S}_r$ to a retargeted motion sequence $\mathcal{Y}_r$:

$$\mathcal{Y}_r = f_\theta(\mathcal{X}, \mathcal{S}_r) = \{\mathbf{Y}_r^{\text{dof}}, \mathbf{Y}_r^{\text{root}}\} \tag{2}$$

where $\mathbf{Y}_r^{\text{dof}} \in \mathbb{R}^{L\times N_{dof}}$ denotes the joint angle trajectories of the robot (e.g., $N_{dof} = 29$ for a Unitree G1 robot), and $\mathbf{Y}_r^{\text{root}} \in \mathbb{R}^{L\times 9}$ denotes the root state, composed of the root translation $\mathbf{p}_r^{\text{root}} \in \mathbb{R}^3$ and a continuous 6D rotation representation $\mathbf{r}_r^{\text{root}} \in \mathbb{R}^6$. Following the standard 6D rotation parameterization, $\mathbf{r}_r^{\text{root}}$ corresponds to the first two columns of the root rotation matrix, from which a valid rotation in SO(3) is recovered via orthonormalization.

Notably, our framework utilizes a reference set as a supervisory signal generated by an offline optimization-based retargeter and refined by physical constraints. By learning to internalize these physically plausible mapping patterns, the neural network f_θ facilitates efficient, real-time inference across diverse embodiments.

4 Methodology

4.1 Initial Trajectory Generation

Before applying our proposed refinement method, we first establish a fundamental geometric mapping. We utilize existing optimization-based retargeting methods (e.g., the inverse kinematics-based General Motion Retargeting (GMR) solver) to generate a set of initial robot trajectories $\mathcal{R}^{init}$. These prior methods focus on minimizing the distance between human joints and their corresponding robot joints, thereby providing a geometrical starting point.

However, because these methods lack explicit awareness of the complex topologies within each specific robot embodiment, $\mathcal{R}^{init}$ frequently suffers from physical violations. Specifically, we observe severe local self-collisions (e.g., the robot's hand penetrating its torso) and hazardous near-contacts. Consequently, these initial

trajectories are highly unsafe for real-world deployment and unsuitable as direct ground-truth labels for neural network training.

4.2 Minimum-Change Refinement

To further improve the physical plausibility of retargeted motions, we apply a lightweight post-refinement procedure to frames exhibiting self-collisions or hazardous near-contacts. Rather than solving a continuous global optimization problem over the entire motion sequence, our method performs a discrete local search over a small subset of collision-related DoFs at each problematic frame, which makes the refinement efficient and stable in practice.

Specifically, for each problematic frame t, we identify a local DoF set $\mathcal{D}_t$ from the kinematic chains associated with the contacting bodies, and evaluate a small set of candidate perturbations for each DoF:

$$\Delta q_{t,d} \in \{0, +\delta, -\delta\}, \quad d \in \mathcal{D}_t, \tag{3}$$

where δ is a small angular step. Candidate updates are selected by prioritizing collision reduction and, when multiple candidates yield similar collision costs, choosing the one with the lowest motion-preserving energy:

$$\mathcal{E}_t = \lambda_C \mathcal{L}_C + \lambda_S \mathcal{L}_S + \lambda_D \mathcal{L}_D + \lambda_M \mathcal{L}_M + \lambda_{\text{dev}} \mathcal{L}_{\text{dev}}. \tag{4}$$

Here, $\mathcal{L}_C$ penalizes self-collisions and overly close body contacts, $\mathcal{L}_S$ preserves semantic directional consistency with the source human pose, $\mathcal{L}_D$ maintains root-relative geometric alignment, $\mathcal{L}_M$ promotes local temporal smoothness, and $\mathcal{L}_{\text{dev}}$ constrains the refined motion to remain close to the initial retargeted trajectory.

This design follows a minimum-change principle: collision reduction is treated as the primary objective, while all remaining terms act as regularizers that prevent unnatural corrections. As a result, the refinement resolves physically problematic local configurations through small kinematic adjustments, while largely preserving the motion semantics, geometric structure, and temporal continuity of the original retargeted sequence. Full objective definitions and implementation details are provided in the supplementary material.

4.3 The MaMoRetarget Network

Although offline refinement produces high-fidelity trajectories with low collision rates and good physical validity, its inherently iterative, sequence-wise optimization process renders it impractical for real-time applications. To bridge this gap, we introduce MaMoRetarget, an end-to-end neural retargeting network that amortizes the intricate geometric and physical priors encoded in the refined dataset $\mathcal{R}^*$. We cast cross-embodiment motion retargeting as a conditional sequence-to-sequence translation task. As shown in Fig. 2, MaMoRetarget consists of three principal modules: a morphology-conditioned human motion encoder, a morphology-aware robot specification encoder, and a cross-attention motion decoder augmented with explicit kinematic constraints.

4.3.1 Morphology-Conditioned Human Motion Encoder. As shown in Fig. 3, the objective of the Human Motion Encoder is to encode structured human motion sequences into spatio-temporal representations that preserve local pose structures, global body context, and temporal dynamics. This module provides a queryable motion memory for subsequent embodied conditional decoding.

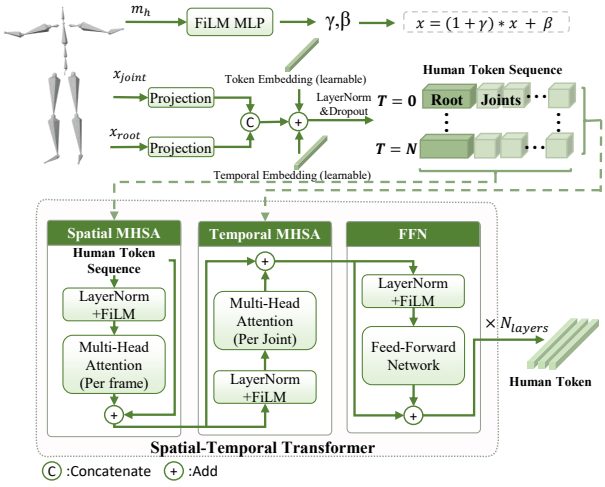


Figure 3: Structure of Human Encoder

Input Representation and Tokenization. Given a human motion clip of length L, the joint-level input is denoted as $X_{\text{joint}} \in \mathbb{R}^{B\times L\times N_h\times D_j}$ and the root input as $X_{\text{root}} \in \mathbb{R}^{B\times L\times D_r}$, where B is the batch size and N_h is the number of human joints. We project both joint and root features into a shared latent space $\mathbb{R}^D$. Each frame is then represented as a sequence of tokens consisting of one root token and N_h joint tokens:

$$H^{(0)} \in \mathbb{R}^{B\times L\times (N_h+1)\times D}. \tag{5}$$

This root/joint decomposition explicitly distinguishes global body displacement from local articulation.

To enrich the initial representation, we add learnable token embeddings E_{tok} and temporal embeddings E_{temp}:

$$\tilde{H}^{(0)} = H^{(0)} + E_{\text{tok}} + E_{\text{temp}}. \tag{6}$$

E_{tok} identifies different body parts and E_{temp} encodes frame order, forming the foundation for spatio-temporal semantic modeling.

Factorized Spatial-Temporal Transformer. We employ a Factorized Spatial-Temporal Transformer to encode human motion. For the ℓ-th block, the input $H^{(\ell-1)}$ is sequentially updated through spatial self-attention, temporal self-attention, and a feed-forward network:

$$H^{(\ell)}_{\text{spa}} = H^{(\ell-1)} + \text{MHSA}_{\text{spa}}(\text{FiLM}(\text{LN}(H^{(\ell-1)}), \mathbf{m}_h)), \tag{7}$$

$$H^{(\ell)}_{\text{tmp}} = H^{(\ell)}_{\text{spa}} + \text{MHSA}_{\text{tmp}}(\text{FiLM}(\text{LN}(H^{(\ell)}_{\text{spa}}), \mathbf{m}_h)), \tag{8}$$

$$H^{(\ell)} = H^{(\ell)}_{\text{tmp}} + \text{FFN}(\text{FiLM}(\text{LN}(H^{(\ell)}_{\text{tmp}}), \mathbf{m}_h)). \tag{9}$$

We adopt a spatial-then-temporal execution order, where spatial attention captures intra-frame dependencies and temporal attention models inter-frame motion evolution. This factorized design better matches the structural prior of skeletal motion while remaining more efficient than global spatio-temporal attention.

Morphology-Conditioned FiLM Modulation. To ensure the encoded representations are explicitly aware of human morphological variations, we introduce Morphology-Conditioned Feature-wise Linear Modulation (FiLM) [23]. Given a human morphology descriptor $\mathbf{m}_h$, a FiLM MLP predicts feature-wise modulation parameters

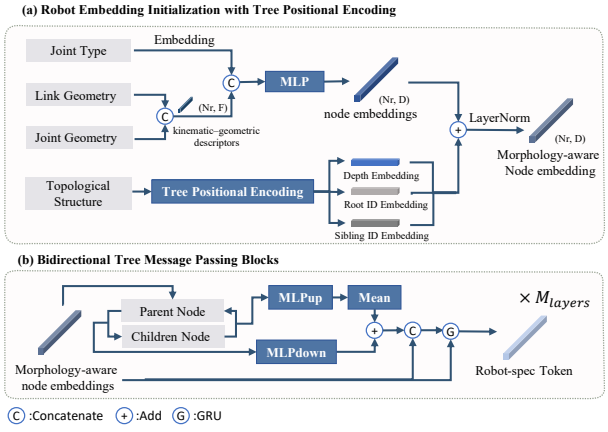


Figure 4: Overview of Robot Encoder.

(γ, β), and transforms the hidden representation h as

$$\mathrm{FiLM}(h, \mathbf{m}_h) = (1 + \gamma) \odot h + \beta. \tag{10}$$

By applying FiLM modulation across attention and FFN layers, the encoder adapts to morphology-dependent statistical variations while preserving motion semantics.

After N_{layers} factorized blocks, the encoder outputs a comprehensive spatio-temporal human token sequence, providing a motion representation that is simultaneously semantic, temporal, and morphology-aware.

4.3.2 Kinematic Tree-Based Robot Specification Encoder. As illustrated in Fig. 4, the robot encoder constructs an embodied representation that explicitly captures local joint attributes, kinematic topology, and cross-joint dependencies. We encode the target morphology into a set of topology-aware robot-specific tokens through two phases: *Robot Embedding Initialization with Tree Positional Encoding* and *Bidirectional Tree Message Passing Blocks*.

(a) Robot Embedding Initialization with Tree Positional Encoding. Given the target robot specification $\mathcal{S}_r$, we represent the target robot as a rooted kinematic tree $\mathcal{T}_r = (\mathcal{V}_r, \mathcal{E}_r)$, where each node corresponds to a robot joint. For each joint $i \in \mathcal{V}_r$, we extract a continuous local descriptor

$$\mathbf{f}_i = \left[\mathbf{a}_i; \mathbf{g}_i; q_i^{\min}; q_i^{\max}; q_i^{\text{rest}}\right], \tag{11}$$

where $\mathbf{a}_i$ is the local joint axis and $\mathbf{g}_i$ denotes the joint/link geometric descriptor. The discrete joint type is mapped to a learnable embedding $\mathbf{e}_i^{\text{type}}$, and the initial node embedding is obtained by

$$\mathbf{h}_i^{(0)} = \mathrm{MLP}\left([\mathbf{f}_i; \mathbf{e}_i^{\text{type}}]\right). \tag{12}$$

To further encode the structural role of each joint, we add three tree positional encodings derived from $\mathcal{T}_r$: the depth index d_i, the root identity r_i, and the sibling index s_i:

$$\tilde{\mathbf{h}}_i^{(0)} = \mathrm{LN}\left(\mathbf{h}_i^{(0)} + \mathbf{e}_{d_i}^{\text{depth}} + \mathbf{e}_{r_i}^{\text{root}} + \mathbf{e}_{s_i}^{\text{sib}}\right). \tag{13}$$

This initialization yields morphology-aware node embeddings that encode both local kinematic-geometric attributes and global structural roles in the robot kinematic tree.

(b) Bidirectional Tree Message Passing Blocks. Building upon these morphology-aware embeddings, we employ Bidirectional Tree Message Passing Blocks to model inter-joint dependencies across the kinematic tree. In the ℓ-th message passing layer, each node maintains a hidden state $\mathbf{h}_i^{(\ell)}$ and receives both bottom-up and top-down structural messages.

For a parent node p, the upward message is computed by aggregating transformed child representations:

$$\mathbf{m}_p^{\uparrow} = \frac{1}{|\mathrm{Child}(p)|} \sum_{c \in \mathrm{Child}(p)} \phi_{\uparrow}\left(\mathbf{h}_c^{(\ell)}\right), \tag{14}$$

while for a non-root node c, the downward message is propagated from its parent:

$$\mathbf{m}_c^{\downarrow} = \phi_{\downarrow}\left(\mathbf{h}_{p(c)}^{(\ell)}\right). \tag{15}$$

We then fuse the bidirectional messages with the current node state:

$$\mathbf{u}_i^{(\ell)} = \left[\mathbf{m}_i^{\uparrow} + \mathbf{m}_i^{\downarrow}; \mathbf{h}_i^{(\ell)}\right], \tag{16}$$

and update the node using a Gated Recurrent Unit (GRU):

$$\hat{\mathbf{h}}_i^{(\ell+1)} = \mathrm{GRU}\left(\mathbf{u}_i^{(\ell)}, \mathbf{h}_i^{(\ell)}\right). \tag{17}$$

Finally, the next-layer representation is obtained through residual connection and normalization:

$$\mathbf{h}_i^{(\ell+1)} = \mathrm{LN}\left(\mathbf{h}_i^{(\ell)} + \mathrm{Dropout}\left(\hat{\mathbf{h}}_i^{(\ell+1)}\right)\right). \tag{18}$$

After stacking M_{layers} bidirectional tree message-passing blocks, the encoder produces final topology-aware robot-spec tokens. Each token captures both the local properties of its associated joint and its structural role within the full robot embodiment, making them suitable as embodiment-conditioned queries for the downstream decoder.

4.3.3 Cross-attention Decoder. Upon obtaining the human motion and robot-spec tokens, the decoder performs the final cross-embodiment motion translation. We implement a cross-attention mechanism that uses the robot-specific tokens as the Query and the human motion tokens as the Key and Value. This enables each robot joint to adaptively retrieve relevant motion semantics from the human representation according to its structural role. Consequently, the model facilitates effective retargeting without relying on rigid, one-to-one correspondences across disparate morphologies.

Formally, let $\mathbf{H} \in \mathbb{R}^{B \times L \times N \times D}$ denote the human motion tokens and $\mathbf{R} \in \mathbb{R}^{N_r \times D}$ denote the robot-specification tokens. At each timestep t, the decoder computes:

$$\mathbf{Z}_t = \mathrm{MHA}(\mathbf{Q} = \mathbf{R}, \mathbf{K} = \mathbf{H}_t, \mathbf{V} = \mathbf{H}_t), \tag{19}$$

Subsequently, the attention outputs $\mathbf{Z}_t$ are processed by a Degree-of-Freedom (DoF) head to predict the robot's joint trajectories, while an independent Root head estimates the robot's root state. To guarantee kinematic feasibility, we explicitly enforce joint limits within the output space. For constrained joints, we employ a tanh-based mapping to bound the predictions within the valid interval $[q_{\min}, q_{\max}]$. For unconstrained joints, the model predicts the motion residuals relative to their rest poses. Through this formulation, the decoder elegantly fuses human motion semantics with robot morphological constraints.

5 Experiments and Analyses

We conduct experiments to evaluate physical plausibility, temporal smoothness, multi-robot joint training, downstream policy learning, and runtime efficiency.

5.1 Experimental Setup

5.1.1 Dataset. We curate a diverse subset of motion sequences from the LAFAN1 dataset [7], ranging from fundamental locomotion (e.g., walking and turning) to highly dynamic and complex skills (e.g., martial arts, kicking, and dancing). The dataset is partitioned into training and testing splits. The training set encompasses the majority of the motion categories, whereas the testing set intentionally excludes actions requiring complex environmental interactions, such as stair climbing and hurdling. To evaluate our cross-embodiment capabilities, we conduct experiments on three distinct humanoid robots: Unitree G1, Booster T1, and Fourier N1.

5.1.2 Baseline. We benchmark our proposed method against three representative retargeting baselines: PHC [19], ProtoMotions (PM) [29], and GMR [3]. PHC and ProtoMotions are specifically designed for SMPL-format motions [18] and thus require an additional format conversion step for comparison, whereas GMR operates directly on raw motion-capture data. To further evaluate the effectiveness of our method, we consider two variants: Ours-R, which denotes motion data obtained from GMR retargeting followed by refinement with our proposed optimization mechanism, and Ours-M, which denotes motion data generated directly by MaMoRetarget.

5.1.3 Configuration. During the minimum-change optimization phase, we set the collision margin to 0.002 m and the angular tolerance to 0.03 rad, conducting the optimization over two epochs. For the MaMoRetarget network architecture, both the Human Encoder and the Robot Encoder feature a latent dimension of 256 and are constructed with four layers. The cross-attention decoder is similarly configured with 4 attention heads. The model is trained using the AdamW optimizer coupled with a cosine decay learning rate schedule. To maximize efficiency, we enable Automatic Mixed Precision (AMP). All training procedures are executed on a server equipped with a single NVIDIA RTX 4090D (24 GB) GPU.

5.1.4 Evaluation Metrics. The goal of motion retargeting is to generate reference trajectories that are suitable for downstream policy learning, enabling the target robot to faithfully imitate human motions. We adopt a progressive evaluation protocol. First, we evaluate the self-collision behavior of retargeted motions in MuJoCo to assess their physical plausibility. Second, we use jerk RMS to measure the temporal smoothness and continuity of the trajectories. Third, we report the mean squared error (MSE) under different numbers of target robots to examine whether MaMoRetarget learns topology-aware knowledge that generalizes across embodiments. Finally, we evaluate the retargeted motions in downstream policy learning. Specifically, we employ BeyondMimic [15] in Isaac Sim to train tracking policies on motions generated by different methods. A retargeted sequence is considered successful if the learned policy can execute the entire motion while keeping the height and orientation deviations of the robot's anchor bodies within predefined thresholds. We therefore use the success rate as the primary evaluation metric. Following [33], we further report the mean global body-part position error (Eg-mpbpe, mm), the mean body-part position error relative to the root (E-mpbpe, mm), and the mean joint rotation error (E-mpjpe, 10^{-3} rad) to evaluate geometric tracking quality.

5.2 Experimental Performance

5.2.1 Physical Plausibility Analysis in MuJoCo. We first replay the retargeted motions in MuJoCo and report the number of collision frames, collision events, and penetration depth to assess their prior physical plausibility. These metrics mainly reflect the geometric validity of the retargeted trajectories as reference motions, rather than their final effectiveness for downstream policy learning. In practice, reinforcement learning can partially compensate for local collision artifacts through collision-aware rewards, whereas overly aggressive collision suppression may distort temporal structure, root–body coordination, or task-relevant motion semantics. Therefore, better collision statistics do not necessarily imply better downstream imitation performance. As will be shown later in Tab. 4, this also helps explain why PHC, despite showing slightly better collision statistics in Tab. 1, underperforms PM in policy learning evaluation. Overall, Tab. 1 shows that our method produces physically more plausible retargeted motions. We thus treat MuJoCo-based collision statistics as complementary diagnostic measures and assess retargeting quality jointly with downstream policy success rates and tracking errors.

In Fig. 5, we present a qualitative comparison of retargeting results across different methods. The rightmost column shows the original motion capture sequence, while the first four columns display the corresponding retargeted motions. Green indicates collision-free frames, whereas red indicates collision frames, with the colliding regions highlighted by circles. The arrows represent the direction and magnitude of the contact force. More results and visual analyses can be found in the supplementary material.

5.2.2 Temporal Smoothness Analysis in MuJoCo. Complementing the collision analysis, we further report Jerk RMS in MuJoCo, defined as the root mean square of the time derivative of joint angular acceleration, to measure the temporal smoothness of the retargeted motions. Whereas collision metrics capture spatial geometric validity, Jerk RMS reflects high-order kinematic smoothness and thus offers a complementary perspective on control-friendliness. As shown in Tab. 2, the minimum-change refined GMR results (Ours-R) do not show a clear advantage over prior methods under this metric, likely because Ours-R still relies on frame-wise optimization. Although such local adjustments help suppress collisions while preserving consistency with the original trajectories, they can also introduce high-frequency jitter. In contrast, MaMoRetarget (Ours-M) achieves the best performance, suggesting that it not only produces physically more plausible motions but also better captures motion priors induced by different robot topologies and physical properties, thereby generating smoother joint trajectories. Similar to collision statistics, Jerk RMS should be viewed as a complementary diagnostic indicator of reference motion quality rather than a direct predictor of downstream imitation performance. We therefore evaluate retargeting quality jointly with collision statistics, downstream policy success rates, and tracking errors.

Table 1: Collision statistics comparison of different retargeting methods on three humanoid embodiments.

	G1			T1			N1		
Methods	**Coll. Frames↓**	**Coll. Count↓**	**Pen. Dist.↓**	**Coll. Frames↓**	**Coll. Count↓**	**Pen. Dist.↓**	**Coll. Frames↓**	**Coll. Count↓**	**Pen. Dist.↓**
PHC	921.4	1741.3	0.0137	1173.9	1770.5	0.01932	120.8	204.2	0.00949
PM	1332.1	4909.1	0.0133	1937.8	3911.5	0.01621	671.1	800.0	**0.00852**
GMR	1144.9	3552.6	0.0133	1895.0	3834.8	0.01608	239.5	332.1	0.00881
Ours-R	**148.5**	**454.8**	**0.0084**	**446.9**	**753.5**	**0.01095**	**24.5**	**42.8**	0.00936
Ours-M	627.8	1836.7	0.0126	1121.7	2292.9	0.01435	63.7	80.0	0.00948

Best results are marked by darker shading and boldface, and second-best results are underlined. "–" indicates unavailable results.

Table 2: Robot jerk RMS statistics of different retargeting methods on three humanoid embodiments.

	G1					T1					N1				
	PHC	**PM**	**GMR**	**Ours-R**	**Ours-M**	**PHC**	**PM**	**GMR**	**Ours-R**	**Ours-M**	**PHC**	**PM**	**GMR**	**Ours-R**	**Ours-M**
min	194.4	196.1	202.1	206.0	**148.1**	209.1	216.9	216.6	226.8	**172.8**	247.0	252.4	252.9	253.0	**212.7**
median	519.2	537.9	521.4	523.1	**394.1**	486.1	488.1	487.9	490.7	**385.1**	559.5	580.8	580.3	580.3	**475.6**
mean	518.0	531.1	520.2	521.7	**391.2**	546.9	529.1	529.1	536.8	**417.7**	629.4	641.4	648.1	647.9	**526.9**
max	856.7	894.2	873.4	873.8	**663.2**	931.0	864.7	865.6	871.8	**702.4**	1024.6	1079.6	1073.3	1073.1	**874.9**

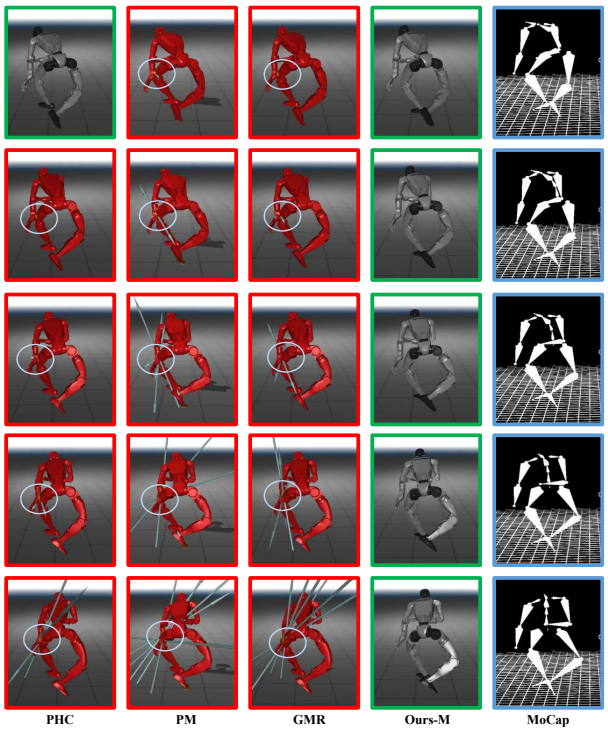


Figure 5: Visualization of retargeting results across different methods.

5.2.3 Ablation on Multi-Robot Joint Training. To evaluate whether MaMoRetarget benefits from multi-robot joint training, we conduct an ablation under one-robot, two-robot, and three-robot training settings, and compare the root and joint MSE on the test set. In general, jointly training on robots with increasingly diverse embodiments introduces larger embodiment-specific variations, which can increase learning difficulty and potentially degrade fitting performance. If the model cannot extract shared structural and physical regularities across robots, the fitting errors would be expected to increase as more robots are included in training. However, as shown in Tab. 3, our method does not degrade with more training robots; instead, it achieves the best performance under the three-robot joint training setting. This suggests that MaMoRetarget can effectively exploit shared topology- and physics-related priors across different humanoid embodiments, rather than merely memorizing robot-specific motion mappings.

Table 3: Cross-training results with different robot combinations.

	Root MSE(10^{-3})↓			Dof MSE(10^{-3})↓		
Train data	**G1**	**T1**	**N1**	**G1**	**T1**	**N1**
G1	2.085	–	–	8.534	–	–
T1	–	8.427	–	–	24.436	–
N1	–	–	1.743	–	–	41.484
G1+T1	0.987	8.631	–	**4.46**	25.769	–
G1+N1	0.862	–	1.472	5.17	–	40.475
G1+T1+N1	**0.717**	**7.716**	**1.42**	4.602	**24.119**	**37.514**

5.2.4 Performance on Policy Learning. This experiment evaluates the effectiveness of both the proposed minimum-change optimization mechanism and MaMoRetarget on RL. Specifically, we employ BeyondMimic in Isaac Sim to train motion imitation policies from retargeted motion data generated by different methods.

Table 4: Per-motion tracking success rate (%) of different retargeting methods on three humanoid embodiments.

		G1					T1					N1				
Motion	**Frames**	**PHC**	**PM**	**GMR**	**Ours-R**	**Ours-M**	**PHC**	**PM**	**GMR**	**Ours-R**	**Ours-M**	**PHC**	**PM**	**GMR**	**Ours-R**	**Ours-M**
aiming1_subject1	7184	94	94	94	**99**	95	38	12	19	**87**	86	0	0	0	0	0
jumps1_subject1	7334	71	75	73	**95**	89	0	0	0	0	**4**	0	0	0	0	0
run2_subject4	7345	68	65	65	67	**80**	0	0	0	0	0	0	0	0	0	0
walk1_subject1	7840	99	99	**100**	**100**	**100**	80	96	**98**	97	96	0	0	0	0	0
walk2_subject1	7146	80	**88**	85	85	85	0	0	0	0	**83**	0	16	**56**	52	52
walk3_subject5	7399	75	86	87	87	**89**	51	46	16	**77**	74	80	0	35	29	**86**

Table 5: Per-motion geometric error comparison of different retargeting methods on Unitree G1.

	aiming1_subject1			jumps1_subject1			run2_subject4			walk1_subject1			walk2_subject1			walk3_subject5		
Methods	**Eg-mpbpe↓**	**E-mpbpe↓**	**E-mpjpe↓**	**Eg-mpbpe↓**	**E-mpbpe↓**	**E-mpjpe↓**	**Eg-mpbpe↓**	**E-mpbpe↓**	**E-mpjpe↓**	**Eg-mpbpe↓**	**E-mpbpe↓**	**E-mpjpe↓**	**Eg-mpbpe↓**	**E-mpbpe↓**	**E-mpjpe↓**	**Eg-mpbpe↓**	**E-mpbpe↓**	**E-mpjpe↓**
PHC	202.1	98.6	130.9	346.2	86.9	168.5	1463.0	84.4	166.2	183.9	99.0	110.4	270.5	94.4	167.6	185.8	67.3	128.6
PM	187.9	77.4	128.0	218.0	69.7	162.2	688.9	68.0	165.4	169.4	83.3	109.2	200.6	**57.5**	159.3	172.0	56.4	128.6
GMR	164.7	**58.2**	125.6	220.2	58.2	162.6	**288.1**	57.3	168.1	148.1	**57.7**	108.6	203.7	60.8	156.6	**148.7**	45.2	128.8
Ours-R	159.0	59.4	**118.1**	225.6	**56.5**	157.1	372.6	56.9	168.5	146.3	60.4	107.8	195.0	60.4	154.0	155.0	43.6	129.6
Ours-M	**155.6**	61.4	123.3	**206.5**	58.5	**147.5**	468.6	**56.1**	**160.0**	**144.5**	60.6	**104.8**	**188.5**	62.0	**150.8**	167.8	**43.3**	**128.0**

Tab. 4 reports the per-motion tracking success rates of different retargeting methods across three humanoid embodiments. Our method achieves the best overall success rate, indicating that its retargeted motions are more suitable as reference trajectories for downstream imitation learning. Tab. 5 further reports the tracking errors on G1, the robot with the highest success rate, where a larger number of successfully executed motions enables a more stable and fair comparison across methods. Our method remains superior under this metric, showing that it not only improves motion executability but also preserves the key geometric characteristics of the reference motion more faithfully.

Taken together, Tab. 4 and Tab. 5 show that success rate and geometric tracking error reflect different aspects of retargeting quality. Tracking error measures spatial fidelity to the reference motion, whereas success rate more directly reflects the physical executability and control-friendliness of the resulting trajectory on the target robot. Therefore, lower geometric error does not necessarily lead to better downstream imitation performance. Our results suggest that the proposed method does not simply optimize frame-wise geometric alignment, but instead achieves a better balance between geometric fidelity and physical executability. Fig. 6 further shows successful hardware execution on Unitree G1 with a policy trained using Ours-M reference motions.

5.2.5 Runtime. Since PHC and PM require additional motion-format conversion for motion-capture inputs, their runtime cannot be fairly compared with end-to-end retargeting methods. We therefore report the computational efficiency of three motion-capture-based pipelines: GMR, Ours-R, and Ours-M. Specifically, GMR runs at 30 FPS, and the refinement module alone operates at 270 FPS. Because Ours-R performs refinement sequentially on top of GMR, its effective throughput is 27 FPS. By comparison, Ours-M achieves **6301** FPS. Despite its substantially higher efficiency, Ours-M still preserves strong retargeting quality, demonstrating that our framework provides an attractive trade-off between motion fidelity and computational cost. This efficiency advantage further suggests its potential for scalable motion generation and latency-sensitive teleoperation scenarios.



Figure 6: Real-world execution on Unitree G1 using policies trained with GMR and Ours-M reference motions.

6 Conclusion

We presented a two-stage framework for morphology-aware human-to-robot motion retargeting across diverse humanoid embodiments. A minimum-change refinement module repairs collision-prone local kinematic chains while preserving motion semantics and temporal continuity, yielding high-quality supervision. Built on these refined trajectories, MaMoRetarget integrates a morphology-conditioned human motion encoder, a kinematic tree-based robot specification encoder, and a cross-attention decoder with explicit joint-limit handling, converting costly offline optimization into a fast feed-forward predictor. Experiments on three humanoid robots demonstrate improved physical plausibility, temporal smoothness, downstream policy learning, and strong multi-robot generalization, while achieving 6301 FPS. These results establish an effective and scalable solution for MoCap-based humanoid motion retargeting.

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